

SEMESTER EXAMINATION- MAY 2023
CLASS: B.TECH. SEMESTER: VITH
PAPER CODE: MACHINE LEARNING-II
PAPER TITLE: BCE-P614

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Discuss the term “Curse of Dimensionality” in reference to machine learning? Explain using simple example.

Question-2: Compare GridSearchCV and RandomizedSearchCV method of hyperparameter search.

Question-3: Explain machine learning and its advantage over traditional programming in brief.

Question-4: Discuss the importance of data visualization in machine learning process. Also name different visualization tools used during ML development.

Question-5: Explain Support Vector Machine (SVM) algorithm with neat & clean diagram in brief.

Question-6: Differentiate between structured and unstructured data with respect to ML development.

Question-7: Explain linear regression in brief using proper example.

Question-8: Explain Cross Validation (CV) technique in brief using diagram. Also State how it avoids sampling error introduced during data sampling process.

Question-9: What do you understand by performance metrics? List different metrics used to evaluate ML model performance during classification task only.

Question-10: Discuss concept of Overfitting & Under-fitting using bias – variance tradeoff.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

Question-1: Explain clustering with its applications in brief. Also explain working principle of any one clustering algorithm in detail .

Question-2: Explain working principle of random forest (RF) algorithm with neat & clean diagram.

Question-3: Explain different process of data preprocessing in brief. Also discuss different methods of handling missing and corrupt values present in dataset.

Question-4: Discuss hyperparameter tuning and its purpose in brief? Explain GridsearchCV method in detail.

Question-5: What do you understand by dimensionality reduction? Also Explain working of principle component analysis (PCA) algorithm using neat and clean diagram.

Question-6: Explain complete life cycle of supervised machine learning in detail using proper process diagram.t

Question-7: Explain concept of Hierarchical clustering in detail with suitable example.

Question-8: discuss Ensemble learning in brief. Also compare Bagging and Boosting method of ensemble learning using proper diagram.